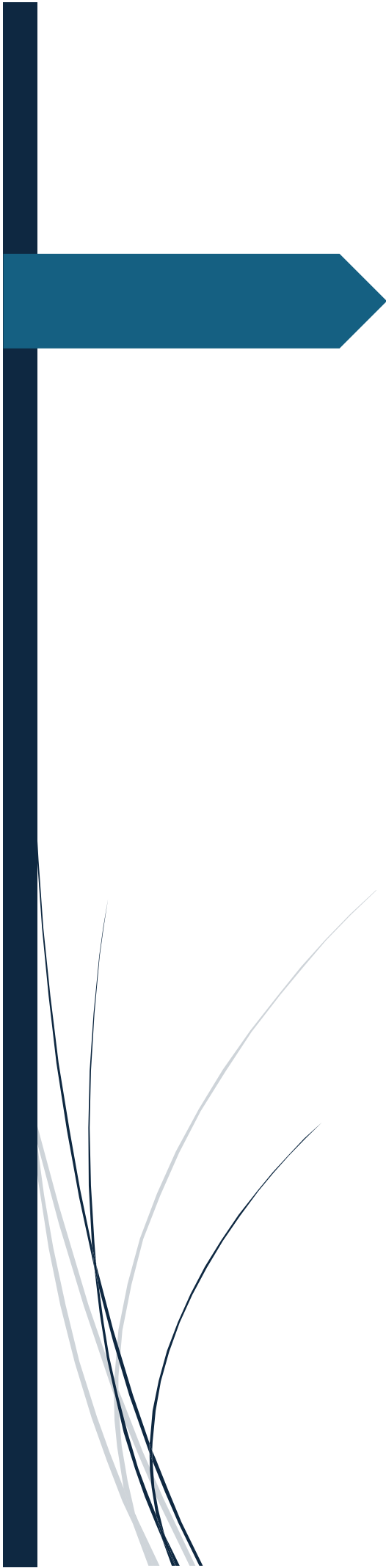




ACTL3301

Models for Risk Management

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We imported daily returns for NVDA and AMD from January 1, 2010, to January 6, 2025. This provides a large sample to make calculations more reliable. In all analyses in this report, X_1 refers to the log returns for NVDA, and X_2 refers to AMD, and the same applies to losses. We fit various copulas in this report and analyse the tail risk and dependence structures.

1: Tail Risk of NVDA vs AMD

We compared the relative tail risk of NVDA and AMD using the value at risk (the VaR) and expected shortfall (ES) at three significance levels. We found the loss corresponding to the log returns X_1 and X_2 where the losses for each is:

$$L_i = 1 - e^{X_i}$$

We found the $VaR_{\alpha}L_i$ at each alpha level for all L_i using the quantile function in R. This function returns the smallest value l such that the cumulative probability is greater than α ($P(L_i \leq l) \geq \alpha$) and the L does not exceed l .

$$VaR_{\alpha}(L) = \inf \{l : P(L > l) \leq 1 - \alpha\}$$

To find the ES_{α} , which is essentially the tail risk of ‘large’ losses, we integrated the $VaR_{\alpha}L_i$ between the confidence level, alpha and 1, then divided by $(1 - \alpha)$ to essentially find the average over the range.

$$ES_{\alpha} = \frac{1}{1 - \alpha} \int_{\alpha}^1 VaR_u(L) du$$

We found the $VaR_{\alpha}L_i$ and ES_{α} at 3 significance levels of 95%, 99% and 99.5%.

α	NVDA $VaR_{\alpha}L$	AMD $VaR_{\alpha}L$	NVDA ES_{α}	AMD ES_{α}
0.95	0.0426	0.0515	0.0626	0.0768
0.99	0.0717	0.0922	0.0963	0.1220
0.995	0.0862	0.1120	0.1140	0.1420

Table 1 – $VaR_{\alpha}L_i$ and ES_{α} for NVDA and AMD

We found that increasing α leads to larger $VaR_{\alpha}L_i$ and ES_{α} . This aligns with expectations, as higher confidence levels reflect more extreme risk quantiles. Further, AMD consistently shows a higher tail risk than NVDA as seen in table 1, evincing that it is the riskier investment solely based on the empirical relative tail risk. Intuitively, it is expected that $VaR_{\alpha}L_i < ES_{\alpha}$, since a higher α level means that the outcome occurring is a rarer event, it naturally corresponds to a larger potential loss. We find that ES_{α} is always significantly greater than $VaR_{\alpha}L_i$, approximately 2% higher at $\alpha = 0.95$ and 3% higher at $\alpha = 0.995$. This shows that as α increases, the difference between $VaR_{\alpha}L_i$ and ES_{α} also increases.

2: Independence Copula – Portfolio Tail Risk

We first fitted the independence copula (X_1, X_2) , which we did by independently generating 10,000 uniform random variables for each margin using `runif()`. We then transform these uniforms into X_1^* and X_2^* using the ECDFs (empirical quantiles) of X_1 and X_2 . We performed this sampling since there is underlying co-dependency in the historical data; hence, we sample part of it to formulate an independent copula. We then computed the simulated losses L_1^* and L_2^* under independence, where the loss formula is the same as in section 1, except this time, using X_1^* and X_2^* instead of X_1 and X_2 .

Since we are working with an equally weighted portfolio, we found each of the losses L_1^* and L_2^* , then combined them by allocating equal weights to find L^* . However, this assumption is a limitation as it is not necessarily the portfolio that minimises risk, it is evident from section 1 that NVDA and AMD have very distinct tail risks meaning an equally weighted portfolio is not ideal. From the fitted independence copula, we calculated the $VaR_{\alpha}L_i^*$ and ES_{α} for the portfolio loss (L^*) using the methods described in section 1 to find VaR and ES .

α	$VaR_{\alpha}(L^*)$	ES_{α}
0.95	0.0333	0.0457
0.99	0.0533	0.0651
0.995	0.0618	0.0726

Table 2 – $VaR_{\alpha}L_i^*$ and ES_{α} for Independence Copula

For an α of 0.99 that there is a 1% chance that the portfolio losses will exceed 5.33% and in the worst 1% of cases, the average loss is 6.51%. This interpretation can be used for the rest of the significance levels. Both the $VaR_{\alpha}L_i^*$ and ES_{α} increase as the significance level increases, which follows the logic from section 1. ES_{α} is also consistently higher in magnitude than VaR. This reflects that the expected losses in the tail are worse than the $VaR_{\alpha}L_i^*$ threshold, hence capturing the tail heaviness present. Across all significance levels, there is a moderate tail risk, suggesting that the best fitting copula for this data accommodates for tail heaviness. However, there is less tail risk compared to the risk with a single asset portfolio in section 1 since we introduce the benefit of diversification with a 2-asset portfolio.

3: Kendall’s Tau and Spearman’s Rho

Concordance and discordance describe the degree and direction of dependence between two random variables, which is useful to understand rank dependence in the context of copulas, which model the joint distribution of variables separately from their marginals. The probabilities of concordance describe the probability that two observations move in the same direction. In contrast, the probability of discordance is the probability that two observations move in opposite directions. Let (X_1, X_2) and $(\tilde{X}, \tilde{Y})$ be two independent pairs of observations that are drawn from the same copula, then the probabilities of concordance and discordance are as follows.

$$P_C = P[(X_1 - \bar{X}_1)(X_2 - \bar{X}_2) > 0] \text{ and } P_D = P[(X_1 - \bar{X}_1)(X_2 - \bar{X}_2) < 0]$$

Once we obtained these values, we calculated Kendall's tau, which is essentially the difference of these probabilities, as seen here. Appendix 1 shows a more in-depth derivation.

$$\begin{aligned}\rho_T(X_1, X_2) &= E \left[\text{sign} \left((X_1 - \bar{X}_1)(X_2 - \bar{X}_2) \right) \right] \\ \rho_T(X_1, X_2) &= P \left((X_1 - \bar{X}_1)(X_2 - \bar{X}_2) > 0 \right) - P \left((X_1 - \bar{X}_1)(X_2 - \bar{X}_2) < 0 \right) \\ \rho_T(X_1, X_2) &= P(\text{Concordance}) - P(\text{Disconcordance})\end{aligned}$$

We found the number of pairs where $(X_1 - \bar{X}_1)(X_2 - \bar{X}_2) > 0$ and $(X_1 - \bar{X}_1)(X_2 - \bar{X}_2) < 0$ to find their proportions, then found the difference of the values. We found a Kendall's tau of 0.443, which conveys that there is a 44.3% greater chance that a randomly chosen pair of observations is concordant (both moving in the same direction) than discordant (moving in opposite directions). The magnitude and sign convey that there is a moderate positive dependence. This correlation technique is especially useful if the relationship between NVDA and AMD's losses is nonlinear and is robust to outliers.

We then found the Spearman's rho, which measures the monotonic dependence and is essentially Pearson's correlation between the distribution functions. To find the distribution functions we transform the loss data into pseudo-observations (U_i) using their ECDFs, which assigns each observation a percentile rank scaled to the interval between 0 and 1. R's 'ecdf(X)(X)' function constructs a step function that increases by $\frac{1}{n}$ at each data point and applies the ECDF to all original x_i , hence U_i is Uniform between (0,1). This process is essential when fitting copulas to empirical data, so that we can then apply the following logic to find the copula.

$$C(u_1, u_2) = P(U_1 \leq u_1, U_2 \leq u_2)$$

Once we find the ECDFs, we find the Pearson's correlation between the ECDFs, which outputs the Spearman's rho.

$$\rho_S(X_1, X_2) = \rho(F_{X_1}(X_1), F_{X_2}(X_2))$$

The correlation was found to be 0.611, which evinces that there is a moderate to strong positive monotonic relationship, higher losses in one asset tend to be associated with higher losses in the other. Spearman's Rho is more sensitive to the overall structure of rank ordering and is more affected by extreme values than Kendall's Tau. The Spearman's rho is considerably higher than Kendall's tau, which may be because rho is more sensitive to extreme values and captures more of a holistic dependence, whereas Kendall's tau has a more direct probabilistic approach.

4: Gaussian Copula

We apply the Gaussian copula model to capture the dependence structure between NVDA and AMD's log returns, X_1 and X_2 , respectively. To estimate $VaR_\alpha L_i$ and ES_α using the estimates for X_1 and X_2 , we first estimate the Pearson's correlation between the observed X_1 and X_2 . With a multivariate Gaussian Distribution, we use the form $X^* = \mu + AZ$. Where μ is the mean and A in this case is the Cholesky factor and Z is a vector of univariate iid standard normal variables.

Using runif(), we generated 10,000 independent, uncorrelated $\widetilde{U}_1$ and $\widetilde{U}_2$, uniformly distributed between 0 and 1. Then, we set Z_i to be the inverse normal of U_i , meaning Z_i is standard normally distributed and iid. We then simulated pseudo-observations for Y_1 and Y_2 , where Y_1 is simply Z_1 . To simulate Y_2 , which is a variable correlated with Z_1 , we used the Cholesky relationship, which gives the following results. The full derivation along with the rest of the methodology for this section is provided in Appendix 2. Here, ρZ_1 introduces the desired correlation and $\sqrt{1 - \rho^2} Z_2$ scales a new independent standard normal so that the standard deviation of Y_2 becomes the correlation.

$$\begin{aligned}\text{Since } \mu = 0: \quad \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} &= A \begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \rho & \sqrt{1 - \rho^2} \end{pmatrix} \begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} \\ Y_1 &= Z_1 \text{ and } Y_2 = \rho Z_1 + \sqrt{1 - \rho^2} Z_2 \quad \text{where } \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \sim N(0, \rho)\end{aligned}$$

Next, we transformed the correlated normal variables Y_1 and Y_2 into uniform variables U_1 and U_2 on the interval [0,1]. Specifically, we found the CDF of both variables, which is the probability that a standard normal variable is less than or equal to z_i and since probability is restricted between 0 and 1, the result is a uniform distribution as seen here: $U_i = \Phi(Y_i)$.

With this, we have simulated a bivariate normal copula $(U_1, U_2) \sim C_\rho^{Ga}$. We applied Sklar's theorem, in which every multivariate cumulative distribution function can be expressed in terms of its marginals and a copula. We mapped Y_1 and Y_2 through empirical quantiles to find samples X_1^* and X_2^* with marginals matching the original dependence, governed by the Gaussian copula C_ρ^{Ga} . We used X_1^* and X_2^* to determine L_1^* and L_2^* , then L^* for an equally weighted portfolio using the method in sections 1-2. We repeated the steps in section 1 to find $VaR_\alpha L^*$ and ES_α .

α	$VaR_\alpha(L^*)$	ES_α
0.95	0.0413	0.0587
0.99	0.0703	0.0868
0.995	0.0821	0.0985

Table 3 – $VaR_\alpha L_i^*$ and ES_α for Gaussian Copula

As usual, higher confidence levels lead to larger estimated losses, but they increase smoother and less extreme than under heavy-tailed copulas which we will explore in the next section. There is also a modest gap between the $Var_{\alpha}L^*$ and ES_{α} which is explained by the Gaussian copula's thin tails. Although the gaussian copula captures linear correlation and symmetric, elliptical dependence, it fails to model tail dependence, this is because its PDF has an exponential with a negative power which means the tails decrease rapidly and thus, it has thick tails. The simulated scatterplot in figure 1 displays moderate dependence near the centre but lacks pronounced tail clustering, reflecting its limitation in capturing joint extreme events. While the Gaussian copula models average dependence well, it underestimates the probability of extreme joint tail events due to its assumption of zero tail dependence, making it unsuitable for modelling systemic market crashes. The plot is symmetric along the $U_1 = U_2$ diagonal, which shows that the dependence is uniform in both tails, however in financial situations, it is more common for assets to crash together than rapidly grow together, this means that when modelling log returns, lower tail dependence is often stronger than upper tail dependence and this copula does not capture this trend. When qualitatively comparing to the empirical scatterplot of U_1 and U_2 in appendix 3, we find that this copula does not exactly resemble the fit in at the tails.

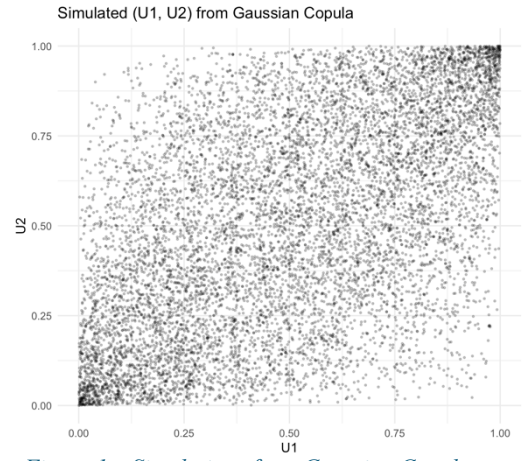


Figure 1 – Simulations from Gaussian Copula

5: T-Copula

We fitted bivariate t-copulas with 3, 10, and 10000 degrees of freedom to model the dependence structure between X_1 and X_2 . Unlike the Gaussian copula, which assumes little to no tail dependence, the t-copula incorporates tail dependence, which is controlled by the degrees of freedom parameter ν . In theory, as ν increases and approaches infinity, the t-copula converges to the Gaussian copula. To simulate the t-copula, we use the variance mixture structure which is an enhanced version of the structure used in section 4: $T = \mu + \sqrt{W}AZ \sim t_{\nu}(\mu, \Sigma = AA^T)$. However, since $\mu = 0$, we are left with $T = \sqrt{W}AZ$

We found $\widetilde{U}_1$ and $\widetilde{U}_2$ just as we did in section 4 by using runif() to generate 10,000 independent, uncorrelated uniforms. Z_i is a bivariate standard normal vector which we found transforming uniforms $\widetilde{U}_1$ and $\widetilde{U}_2$, using the inverse normal function: $Z_i = \phi^{-1}(\widetilde{U}_i)$. A is the Cholesky decomposition of the estimated correlation matrix same as section 4, and W is the independent inverse gamma random variable with shape and rate parameter, $\frac{\nu}{2}$. By multiplying $\sqrt{W}$, we scale/mix the gaussian vector, this process introduces heavier tails and generates a t-distributed vector.

After finding T_1 and T_2 using $T = \sqrt{W}AZ$, we find $F_{t,\nu}(T_i)$ which is the CDF of the t distribution (use pt in R) with ν degrees of freedom which becomes the uniform pseudo-observations U_1 and U_2 . Specifically: $U_i = F_{t,\nu}(T_i)$ and $(U_1, U_2) \sim C_t^{\nu}$. These $(U_1, U_2) \sim C_t^{\nu}$ are realisations of the t-copula with the parameter ν . Since we are completing an analysis with three cases of degrees of freedom, we generate separate samples for $\nu = 3, 10, 10000$ to compare how increasing ν affects the dependence structure. We applied Sklar's theorem as we did in section 4, and we transformed the pseudo-observations for U_1 and U_2 back to the scale of the original losses, which is done using the empirical quantile function of the observed data.

$$X_i^* = \hat{F}_i^{-1}(U_i)$$

$\hat{F}_i^{-1}(U_i)$ is the empirical inverse CDF of the observed log-returns for NVDA and AMD. This method preserves the original marginal distributions and imposes dependence through the copula. Given the transformed samples X_1^* and X_2^* , we found the losses for each asset as we did in section 1, then, we found the loss of the equally weighted portfolio. We then calculated the $Var_{\alpha}L_i^*$ and ES_{α} of the losses in an equally weighted portfolio using the same method as section 1 at all the significance levels and degrees of freedom, which we did by using a loop function in R.

α	ν	$Var_{\alpha}(L^*)$	ES_{α}
0.95	3	0.0398	0.0599
0.99	3	0.0711	0.0937
0.995	3	0.0824	0.1110
0.95	10	0.0427	0.0642
0.99	10	0.0742	0.1010
0.995	10	0.0922	0.1200
0.95	10000	0.0408	0.0598
0.99	10000	0.0710	0.0919
0.995	10000	0.0848	0.1060

Table 4 – $Var_{\alpha}L_i^*$ and ES_{α} for t-Copula

These results portray the impact of tail dependence, since the smaller degrees of freedom correspond to stronger tail dependence, therefore a higher joint tail risk, which is reflected in the heavier clustering at the tails in the copula scatter plots and increased $Var_{\alpha}L_i^*$ and ES_{α} estimates. Additionally, to compare to section 4, we find that the $Var_{\alpha}L_i^*$ and ES_{α} with the t-copulas are generically higher than those of the Gaussian copula. This is intuitive because the t-copula exhibits heavier tails and stronger tail dependence, meaning that extreme losses in one asset are more likely to be accompanied by extreme losses in the other. Whilst we expect that as the degrees of freedom increases, the tail risk decreases, our results present slight anomalies at some significance levels, possibly since we only sampled 10,000 values, limiting reliability.

Figure 2 displays simulated samples from the t-copulas and reveals how tail behaviour changes with ν yet is limited by its qualitative nature which usually is less reliable due to the lack of numerical backing up. With lower ν values, tail dependence appears to be more pronounced and is most visible in the $\nu = 3$ case, where there is strong clustering in the lower-left and upper-right corners, evincing a high chance of simultaneous extreme events. In contrast, the $\nu = 10,000$ has less tail clustering and a smoother, elliptical shape. The $\nu = 10$ case falls in between, as it moderately captures co-

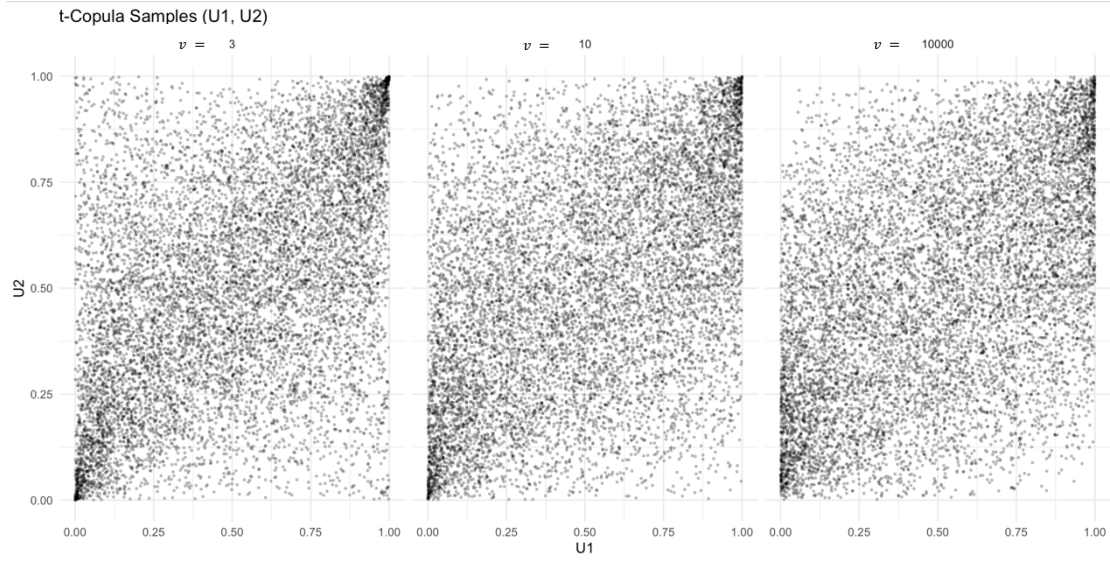


Figure 2 – Simulations from t Copula

extremes while still maintaining symmetry around the diagonal, best resembling the uniforms of the empirical copula in appendix 3 followed by the copula with $\nu = 10,000$. These shape differences are critical in risk modelling, during a market crash, when assets often fall together, the t-copula with $\nu = 10,000$ may underestimate risk, whilst $\nu = 3$ may overestimate the risk. The copula with 10 degrees of freedom is a good ‘in-between’ as it has some heavy tails yet is not too conservative.

6: Comonotonicity and Countermonotonicity Copula

To understand how extreme dependence structures affect portfolio risk, we simulated from two extreme copulas being the comonotonicity copula, which has perfect positive dependence the two and the counter-monotonicity copula, which has perfect negative dependence. These serve as theoretical bounds for dependence between assets.

To generate a comonotonic copula, we simulated 10,000 uniform random variables U using `runif()`, then set both the marginal uniform random variables to come from this Uniform $[0,1]$ distribution to reflect comonotonicity. They are perfectly positively dependent as $U_1 = U_2 = U$. We then transform these using each asset’s empirical quantile function $F_i^{-1}(U)$ to get the simulated log-returns, X_i^* . The X_i^* is mapped to the losses L_1^* and L_2^* , as described in section 1, then an equally weighted portfolio is formed. Using the portfolio loss L^* , we find the $Var_{\alpha} L_i^*$ and ES_{α} as the methodology is explained in section 1.

To generate a counter-monotonic copula, we again used the simulated uniform random variable U_1 and set $U_2 = 1 - U_1$. This ensures that when one asset is in loss, the other tends to gain. Similarly to the comonotonic copula’s process, we transform this using the empirical quantile functions, thus computing the corresponding losses for each asset, which we then use to find the equally weighted portfolio’s loss. We then also find the $Var_{\alpha} L_i^*$ and ES_{α} using the same process.

α	Comonotonic $Var_{\alpha}(L^*)$	Comonotonic ES_{α}	Countermonotonic $Var_{\alpha}(L^*)$	Countermonotonic ES_{α}
0.95	0.0480	0.0692	0.00232	0.00395
0.99	0.0827	0.1040	0.00622	0.00662
0.995	0.0980	0.1190	0.00652	0.00689

Table 5 – $Var_{\alpha} L_i^*$ and ES_{α} for comonotonic and countermonotonic copulas

The $Var_{\alpha} L_i^*$ and ES_{α} for the comonotonic copula shows that there is moderate risk in this portfolio, this is reasonable considering that there are no diversification benefits, as both portfolios move in the same direction monotonically. It suggests that if one stock performs poorly, due to perfect correlation, the other also likely performs poorly. The results for the $Var_{\alpha} L_i^*$ and ES_{α} for the comonotonic copula show that there is significantly more tail risk in this portfolio compared to the countermonotonic case, on average has approximately 16 times more tail risk. This likely results from the diversification benefits in the countermonotonic case as both assets in the portfolio move in opposite directions, suggesting that if one stock performs poorly, the other stock will move in the opposite direction, therefore, the portfolio will not have as high a loss. Both these copulas are unrealistic for our financial case.

7: Comparison of Copulas

To assess the conditional probability of joint losses under different copulas, we estimate $P(F_1(X_1) > \alpha \mid F_2(X_2) > \alpha)$, which becomes $P(U_1 > \alpha \mid U_2 > \alpha)$. Using the copulas from previous sections, we simulate U_i from the CDFs where $F_i(X_i) = U_i$. The tail event $F_1(X_1) > \alpha$ corresponds to the upper tail of the distribution, and we are interested in the proportion of points where both U_1 and U_2 , exceed α , conditional on U_2 being greater than α .

Alpha	Empirical	Independence	Gaussian	Comonotonic	Countermonotonic	t ($\nu = 3$)	t ($\nu = 10$)	t ($\nu = 10000$)
0.95	0.3093	0.0523	0.275	1	0	0.383	0.331	0.333
0.99	0.1795	0.0377	0.143	1	0	0.323	0.339	0.175
0.995	0.150	0.0172	0.118	1	0	0.273	0.283	0.159

Table 4 – $P(F_1(X_1) > \alpha \mid F_2(X_2) > \alpha)$ for all copulas

Copula	Analysis
Independence	Conditional probabilities are close to $1 - \alpha$ especially for smaller α values, this reflects the lack of dependence and matches the theoretical expectation that tail events occur independently.
Gaussian	The probabilities exceed independence values, showing mild upper tail dependence, which is caused by positive correlation, however they are consistently less than the empirical
Comonotonic	Portrays perfect positive dependence, hence conditional probability is always 1, this means that if one variable is in the upper tail, the other always is and the conditional probability is far too high compared to empirical.
Counter monotonic	Has perfect negative dependence, hence, conditional probability is always zero, indicating the two variables cannot be jointly extreme in the upper tail, but is overly optimistic and fails to capture important risk.
T-Copula	Display substantial upper tail dependence, however, the dependency is stronger at lower degrees of freedom (heavier tails). The $\nu = 10,000$ copula has conditional probabilities closest to the empirical, meaning it best captures the overall conditional loss trends which are useful when examining the joint loss of a portfolio.

We then also compared the copulas we generated earlier.

Copula Type	Tail Dependence	Strengths	Weaknesses
Independence	None	Simple baseline	Underestimates joint tail risk
Gaussian	Weak	Captures linear correlation	Fails to capture tail dependence, can understate risk
Comonotonic	Perfect	Captures maximum joint tail risk ('worst' case)	Too extreme, unrealistic and pessimistic for most real data and overestimates risk
Counter monotonic	Negative	Captures perfect negative dependence ('best' case)	Not appropriate for joint tail modelling as it underestimates risk and is too optimistic.
t-Copula ($\nu = 3$)	Strong	Realistic tail dependence; captures joint extremes	Heavy tails, flexible, only use for strong tail risk
t-Copula ($\nu = 10$)	Moderate	Trade-off between tail dependence and normality	Less extreme than $\nu = 3$
t-Copula ($\nu = 10000$)	Approximately Gaussian	Has some tail dependence, less underestimation than Gaussian.	Weak tail dependence which may not work for financial scenarios

The comparison of $Var_{\alpha}L_i^*$ and ES_{α} across copulas reveals insights into how different dependence structures impact tail risk, this is simply visualised in the graph in Appendix 4 and 5. The empirical $Var_{\alpha}L_i^*$ and ES_{α} serve as the benchmark, as they reflect actual joint behaviour observed in historical data. It is seen that both the independence and Gaussian copulas slightly underestimate tail risk ($Var_{\alpha}L_i^*$ and ES_{α}) across all α levels, with the Gaussian performing marginally better due to its ability to capture some linear dependence. The t-copulas, especially with higher degrees of freedom, produce estimates closer to the empirical values at higher confidence levels, highlighting their strength in modelling moderate to strong tail dependence. The pessimistic comonotonic copula consistently yields the highest $Var_{\alpha}L_i^*$ and ES_{α} , reflecting worst-case risk where both assets suffer losses simultaneously. Conversely, the counter-monotonic copula is overly optimistic, due to its assumption that one asset gains whenever the other loses, hence, severely underestimates risk, especially at higher confidence levels. This is unrealistic, as NVDA and AMD historically have a positive correlation and operate in the same industry, hence they do not form a diversified portfolio.

We decide that the t copula with 10,000 degrees of freedom best resembles the tail risk and conditional loss probability. The conditional probability for this copula is the closest to that of the empirical and, on average across α , the $Var_{\alpha}L_i$ and ES_{α} of this copula is closest to that of the empirical (Appendix 4 and 5). Further, it is slightly more conservative than the Gaussian model, suitable for this financial context. Although it is the second-best fitting copula according to the clustering in the scatterplots in Figure 2 (Section 5), that measure is slightly less reliable as it is qualitative, compared to the rest of the quantitative measures, which all support this copula's suitability.

8: Archimedean Copulas

Using the BiCopSelect() function from the VineCopula package in R, we found the best-fitting Archimedean copula. This function evaluates multiple Archimedean copula families and selects the best fitting model by maximising the log-likelihood based on the joint pseudo-observations of the two margins. As we did when creating other copulas, we transformed the marginal distributions into a uniform [0,1] scale using their ECDFs with the 'ecdf(X)(X)' function. This removes the influence of the marginal behaviour and separates the dependence structure. We used these observations in the BiCopSelect() function, which found the best copula from the top 10 most common copulas. We found that the optimal copula is the Survival BB1 (SBB1) copula (BB1 copula rotated 180 degrees). This is a 2-parameter Archimedean copula useful for capturing asymmetric tail dependence. The tail dependence is: $\lambda_U = 2^{-\frac{1}{\delta\theta}}$ and $\lambda_L = 2 - 2^{\frac{1}{\delta\theta}}$

The estimated parameters are $\theta = 0.098$ and $\delta = 1.672$, which control the strength and the shape of dependence, respectively. The lower tail dependence coefficient is approximately 0.486, suggesting a high probability of co-extreme losses between NVDA and AMD, specifically, it indicates a 48.6% chance of extremely low log returns in NVDA when AMD experiences very low log returns. Contrastingly, the upper tail dependence is only approximately 0.0148. This indicates very little co-occurrence of extreme gains. This asymmetry aligns with the copula's strengths and in modelling real-world financial behaviour, as markets tend to drop together more frequently than rise together. Beyond the extremes, the body of the joint distribution reveals that the bulk of the returns lie in the centre, showing a relatively strong positive dependence in

moderate market conditions. This suggests that during normal periods, NVDA and AMD exhibit mild co-movement, likely due to shared industry exposure. Whilst the tail asymmetry highlights asymmetric risk, the body behaviour reflects a smoother, more linear correlation. This body-tail contrast explains the value of using copulas rather than simple correlation or other traditional metrics, which may capture the co-movement well in the centre, but they fail to reflect the nonlinear dependencies and asymmetries that dominate the tails. Thus, the SBB1 copula, which captures the joint downside risk makes it a strong candidate for modelling extreme co-movements between NVDA and AMD.

Using BiCopSim(), we simulated 10,000 pairs of (U_1, U_2) samples from the SBB1 copula, then transformed them into simulated log returns and then losses as we did previously. The unconditional distribution of L_1 was obtained directly from all simulated values, and the conditional distribution of $L_1 | L_2 > VaR_{\alpha}(L_2)$ was computed by filtering for cases where the simulated loss in AMD exceeds $VaR_{0.99}(L_2)$, then found the corresponding values of NVDA's L_1 . To compare the conditional and unconditional distributions of loss L_1 , we created histograms (Figure 3) where $\alpha = 0.99$. The histograms for $\alpha = 0.95, 0.995$ can be found in Appendix 6, and as α increases, the conditional losses increase (moves to the right).

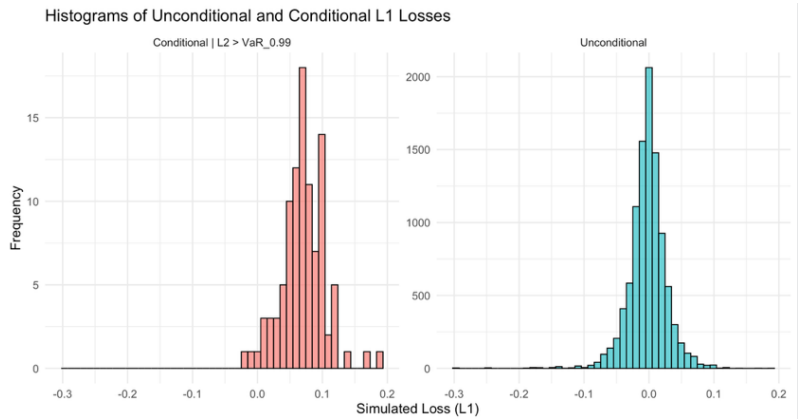


Figure 3 – Conditional vs Unconditional Loss Histograms

The unconditional distribution displays a symmetric shape centred around zero with moderate tails, indicating that small gains and losses are equally likely, while extreme events remain possible but infrequent. In contrast, the conditional distribution shows a shift as the body of the distribution is shifted rightward, with significantly reduced left-tail mass. This suggests that when AMD experiences an extreme loss, NVDA tends to experience smaller losses or even gains, implying asymmetric dependence as aforementioned. The conditional distribution exhibits lower variance, higher skewness, and a positive conditional mean, while the unconditional distribution remains more dispersed but centred at a loss of 0.

The SBB1 copula is the optimal copula as it has a significant advantage over the t-copula with $\nu = 10,000$ due to its ability to model asymmetric tail dependence. The t-copula assumes symmetric dependence in both tails, implying that extreme co-gains are just as likely as extreme co-losses, which does not align with NVDA and AMD's empirical behaviour. There is a strong lower tail dependence and a very weak upper tail dependence, indicating that the assets are far more likely to crash together than to peak together. The t-copula cannot reflect this asymmetry, as it has equal tail weights, whereas the SBB1 copula accommodates this imbalance, making it more effective for risk management. Further, the AIC and BIC of the SBB1 copula is slightly lower than that of a t-copula as seen in Appendix 9 which provides more justification for its suitability.

9: Optimal Portfolio Weights

To identify an optimal weight allocation between NVDA and AMD for a portfolio, we simulated 10,000 joint observations from the fitted SBB1 copula in section 8 and transformed the simulated returns into losses. Unlike previous sections with an equally weighted portfolio, we varied the weight on NVDA from 0 to 1 in increments of 0.0001, computing risk and performance metrics at each level. As discussed in section 2, an equally weighted portfolio is likely not appropriate for this data as the risk tails of NVDA and AMD are very different as seen in section 1. We assessed the expected shortfall at the 99% confidence level and the Sharpe ratio. We used $\alpha = 0.99$ to have higher confidence in our recommendations compared to that with $\alpha = 0.95$. Since we want to analyse the extreme tail risk, we used ES_{α} as it provides a more comprehensive view compared to $VaR_{\alpha}L_i^*$ which only indicates a threshold without accounting for the magnitude of losses beyond it. Specifically, it is useful in this financial context since catastrophic tail events like joint crashes in NVDA and AMD are of major concern. We also used the Sharpe Ratio which is the magnitude of the expected portfolio loss divided by its standard deviation ($\frac{\mu}{\sigma}$). This ratio helps identify weights that achieve the best trade-off between expected return and volatility, reflecting the portfolio's efficiency under normal market conditions.

These two metrics serve complementary purposes as ES_{α} they focus on downside tail risk and are widely used for regulatory and capital adequacy purposes, whilst the Sharpe ratio is valuable from a long-term investment performance perspective. The weight that minimises the ES_{α} was found to be 0.983, and the weight that maximises the Sharpe ratio is 0.979. Both metrics heavily favour NVDA, allocating it close to the full portfolio weight. This suggests that NVDA is a safer investment compared to AMD. To balance the strengths of both benefits, we recommend a portfolio where NVDA is allocated the mean weight of the two metrics, which works out to be 0.981. Although this weight may limit diversification benefits from a more equally allocated portfolio, investing more heavily in NVDA is safer as it helps reduce tail risk by protecting against extreme losses via a low ES_{α} and maintains strong overall performance via a high Sharpe Ratio.

Conclusion

We analysed various copulas to fit the log returns of a portfolio with NVDA and AMD, and before fitting an Archimedean copula, we found the t copula with $\nu = 10,000$ to best fit the empirical data. After fitting the Archimedean copula, we found the SBB1 copula as the best fit as it accounted for the asymmetric tails. We also found that due to the significant difference in the risk, a portfolio with heavily NVDA is optimal.

Appendix

Appendix 1

$$\rho_T(X, Y) = E \left[\text{sign} \left((X - \tilde{X})(Y - \tilde{Y}) \right) \right]$$

$$\rho_T(X, Y) = 1 \times P \left((X - \tilde{X})(Y - \tilde{Y}) > 0 \right) + 0 \times P \left((X - \tilde{X})(Y - \tilde{Y}) = 0 \right) - 1 \times P \left((X - \tilde{X})(Y - \tilde{Y}) < 0 \right)$$

$$\rho_T(X, Y) = P \left((X - \tilde{X})(Y - \tilde{Y}) > 0 \right) - P \left((X - \tilde{X})(Y - \tilde{Y}) < 0 \right)$$

$$\rho_T(X, Y) = P(\text{Concordance}) - P(\text{Disconcordance})$$

Appendix 2

Step 1: Simulate multivariate Normal $N(0, \rho)$

Generate $\widetilde{U}_1, \widetilde{U}_2$ iid $\sim U(0,1)$

$$\text{Set } \begin{cases} Z_1 = \phi^{-1}(\widetilde{U}_1) \\ Z_2 = \phi^{-1}(\widetilde{U}_2) \end{cases} \quad Z_1, Z_2 \text{ iid } \sim N(0,1)$$

Define the Cholesky factor

$$\text{Set } \begin{cases} Y_1 = Z_1 \\ Y_2 = \rho Z_1 + \sqrt{1 - \rho^2} Z_2 \end{cases}$$
$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} = A \begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \rho & \sqrt{1 - \rho^2} \end{pmatrix} \begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix}$$

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \sim N(0, \rho) \text{ where } \rho = AA^T = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$$

Step 2: Simulate copula of bivariate normal $(U_1, U_2) \sim C_\rho^{Ga}$

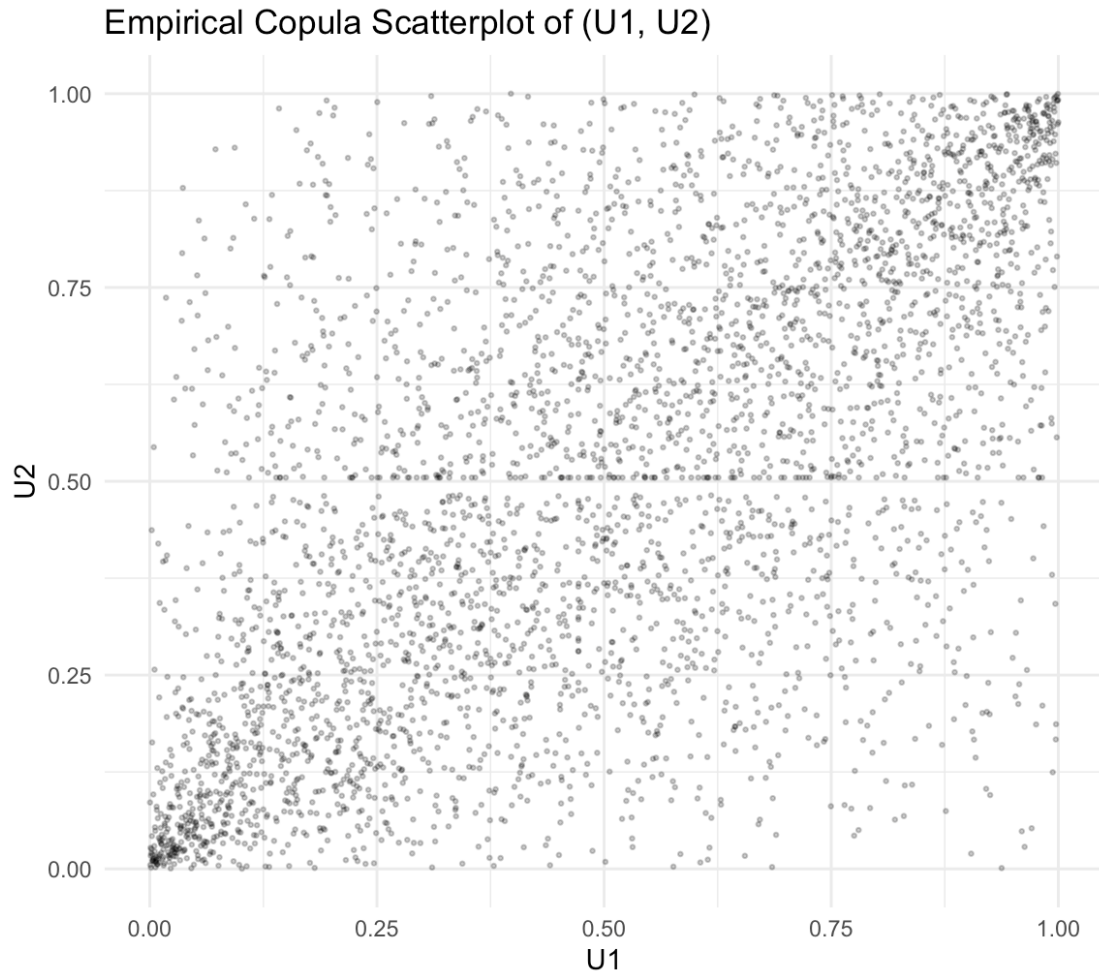
$$\text{Set } \begin{cases} U_1 = \phi(Y_1) \\ U_2 = \phi(Y_2) \end{cases} \text{ where } \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \sim N(0, \rho)$$

$$U_1, U_2 \text{ identical } \sim U(0,1), \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \sim C_\rho^{Ga}$$

From a definition of a copula, we see that

$$C_\rho^{Ga}(u) = \phi_\rho(\phi^{-1}(u_1), \dots, \phi^{-1}(u_d)), \text{ where } \phi^{-1}(u_1) \text{ are marginals}$$

Appendix 3

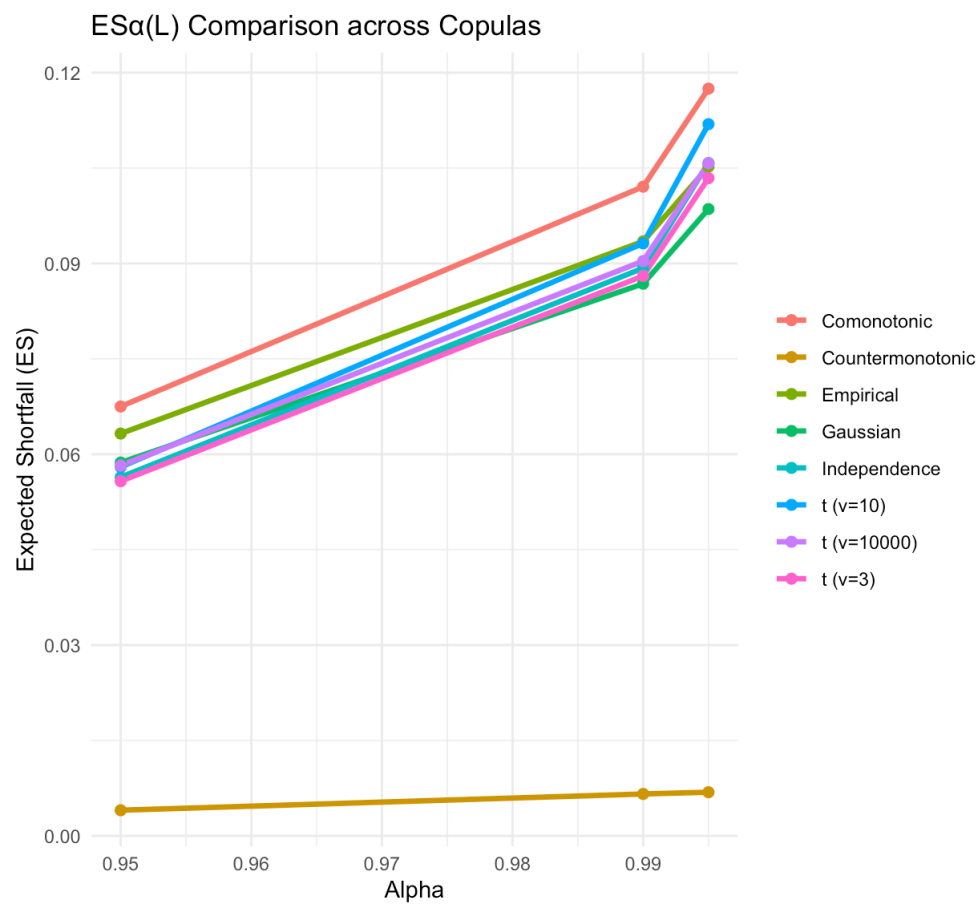
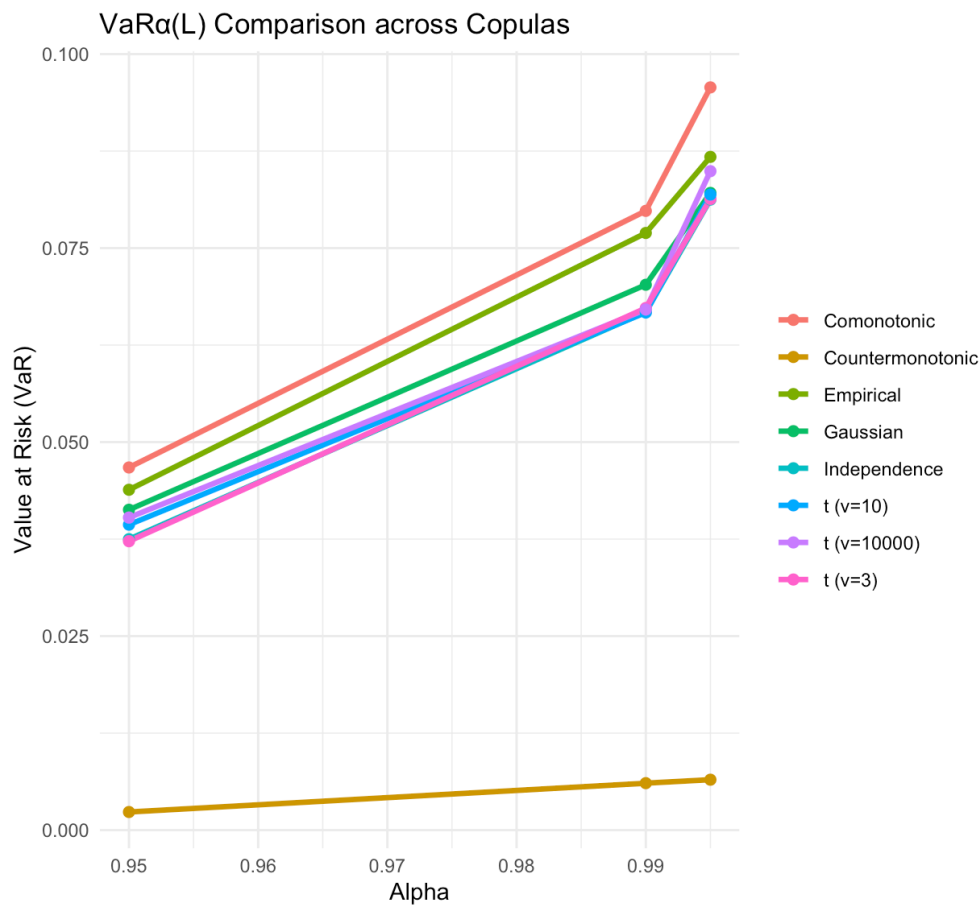


Appendix 4

Combination of all VaR and ES values for all copulas

α	$VaR_{\alpha}(L^*)$	ES_{α}	Copula
0.95	0.0439	0.0633	Empirical
0.99	0.0769	0.0935	Empirical
0.995	0.0868	0.1050	Empirical
0.95	0.0394	0.0556	Independence
0.99	0.0644	0.0836	Independence
0.995	0.0762	0.0981	Independence
0.95	0.0413	0.0587	Gaussian
0.99	0.0703	0.0868	Gaussian
0.995	0.0821	0.0985	Gaussian
0.95	0.0398	0.0599	t (v=3)
0.99	0.0711	0.0937	t (v=3)
0.995	0.0824	0.1110	t (v=3)
0.95	0.0427	0.0642	t (v=10)
0.99	0.0742	0.1010	t (v=10)
0.995	0.0922	0.1200	t (v=10)
0.95	0.0408	0.0598	t (v=10000)
0.99	0.0710	0.0919	t (v=10000)
0.995	0.0848	0.1060	t (v=10000)
0.95	0.0467	0.0675	Comonotonic
0.99	0.0798	0.1020	Comonotonic
0.995	0.0957	0.1170	Comonotonic
0.95	0.00233	0.00403	Countermonotonic
0.99	0.00605	0.00659	Countermonotonic
0.995	0.00650	0.00686	Countermonotonic

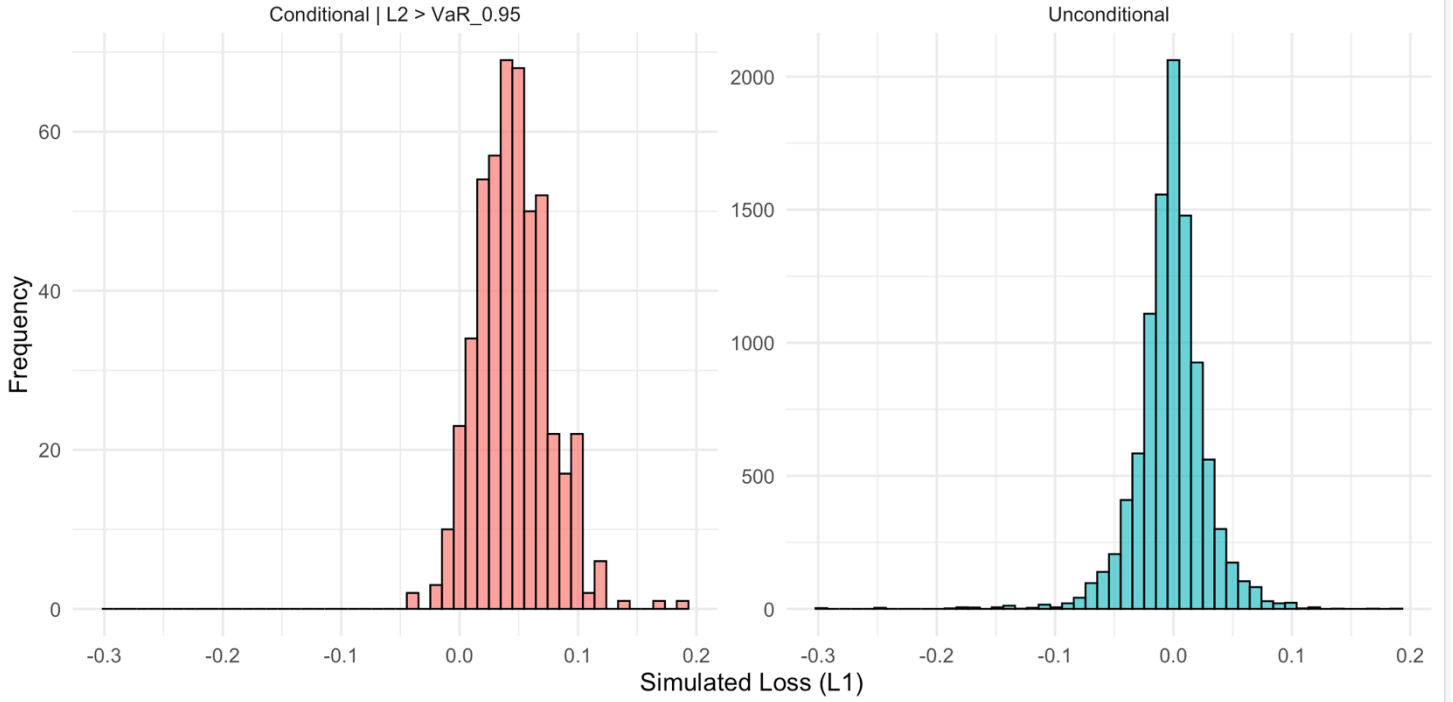
Appendix 5



Appendix 6

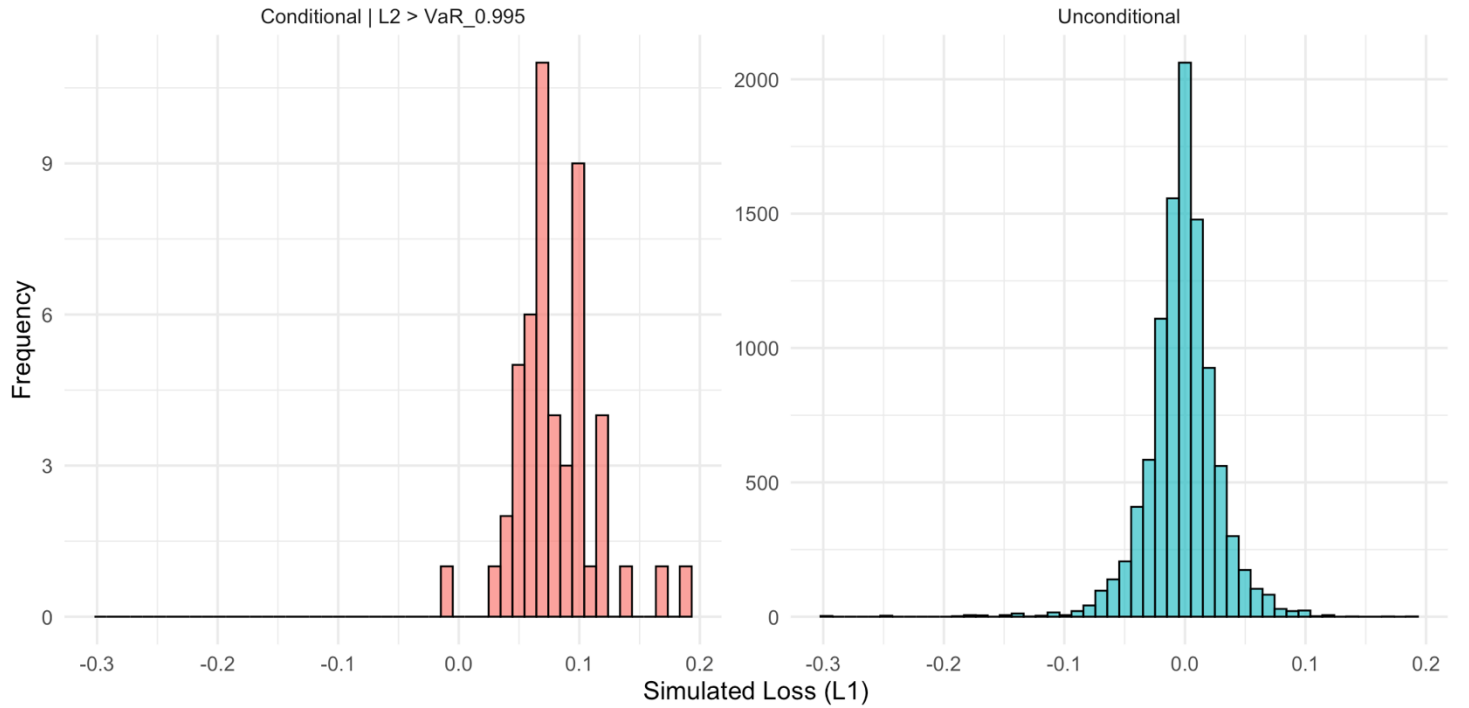
Alpha = 0.95

Histograms of Unconditional and Conditional L1 Losses



Alpha = 0.995

Histograms of Unconditional and Conditional L1 Losses



Appendix 7

Copula

Definition $C(u, v; \theta, \delta)$, ($\bar{u} = 1 - u$)

Tail Dependence

BB1

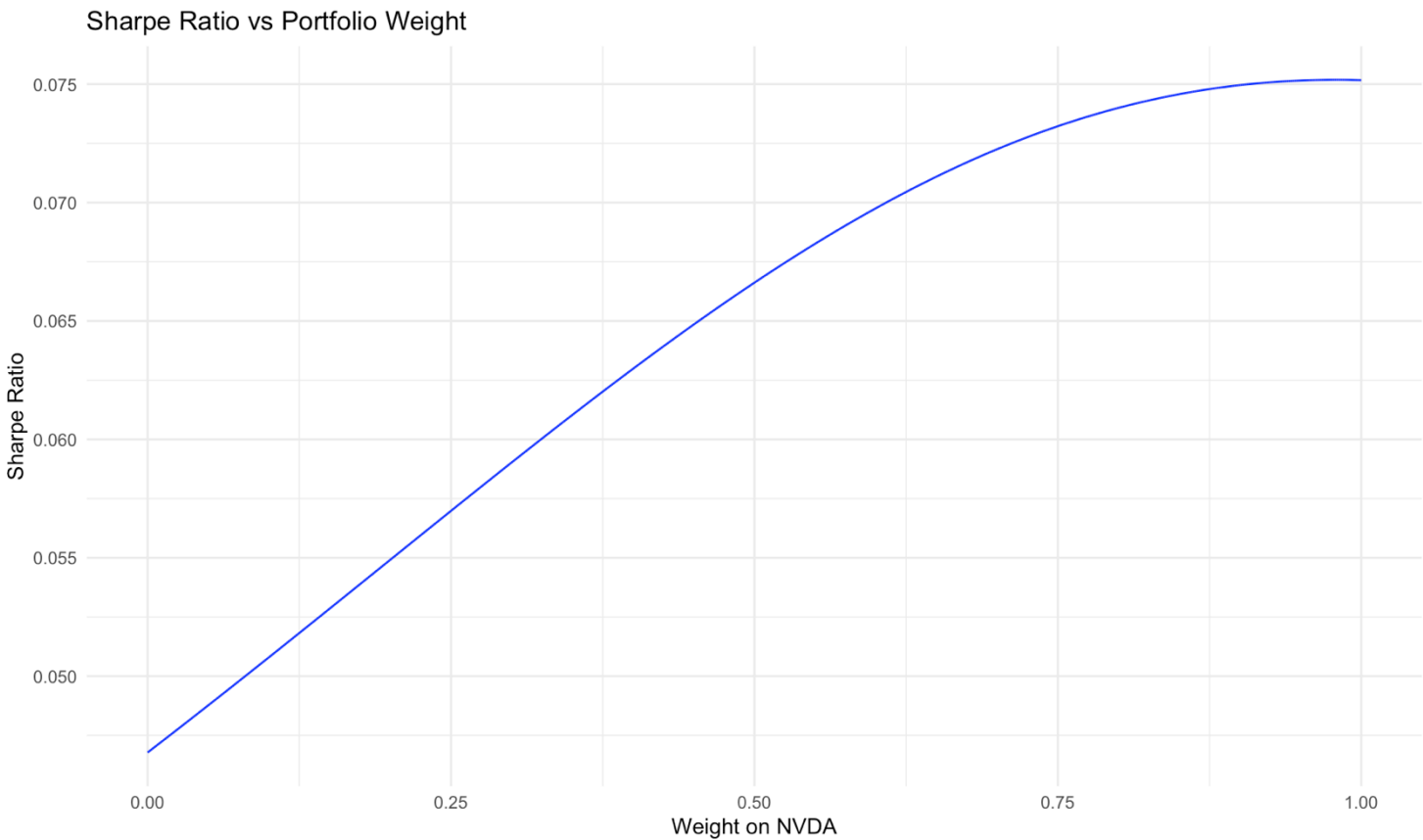
$$\left[1 + ((u^{-\theta} - 1)^\delta + (v^{-\theta} - 1)^\delta)^{1/\delta}\right]^{-1/\theta}$$

$$\lambda_L = 2^{-1/(\delta\theta)}$$

$$\lambda_U = 2 - 2^{1/\delta}$$

Since we use the survival BB1 copula, we find the survival copula formula using $1 - u + v + C(1 - u, 1 - v)$ hence the upper and lower upper and tail dependence swaps.

Appendix 8



Appendix 9

t-Copula Fit Summary

Category	Metric	Value
Family	Name	t
	No	2
Parameters	par (correlation)	0.64
	par2 (degrees of freedom)	5.72
Dependence	Kendall's tau	0.44
	Empirical tau	0.44 ($p < 0.01$)
	Upper Tail Dependence	0.26
	Lower Tail Dependence	0.26
Fit Statistics	Log-likelihood	999.64
	AIC	-1995.27
	BIC	-1982.75

Survival BB1 Copula Fit Summary

Category	Metric	Value
Family	Name	Survival BB1
	No	17
Parameters	par	0.10
	par2	1.67
Dependence	Kendall's tau	0.43
	Empirical tau	0.44 ($p < 0.01$)
	Upper Tail Dependence	0.01
	Lower Tail Dependence	0.49
Fit Statistics	Log-likelihood	1003.69
	AIC	-2003.38
	BIC	-1990.86

AI Appendix

- How to generate uncorrelated uniform variables using `runif()`.
- How to simulate from a t-copula using inverse gamma + Cholesky + standard normal.
- Can `sample()` output continuous values (for X1 and X2 resampling)?
- How to compute the Cholesky decomposition of a correlation matrix (`chol()`).
- How to simulate correlated normals with the Cholesky approach.
- How to transform normal/t values to uniforms using `pnorm()` or `pt()`.
- Misuse of `ecdf(X)` as a function vs correct use of `rank()` for uniforms.
- How to compute Pearson correlation with `cor(X1, X2, method = "pearson")`.
- How to induce correlation in normal variables ($Y2 = \rho Z1 + \sqrt{(1-\rho^2)}Z2$).
- How to apply Sklar's theorem using empirical quantiles (`quantile(..., type = 1)`).
- How to compute VaR using `quantile(losses, probs = 1 -  $\alpha$ )`.
- How to compute ES using `mean(losses[losses > VaR])`.
- How to set a random seed with `set.seed(...)` for reproducibility.
- How to loop over multiple v values (3, 10, 10000) for t-copula simulation.
- How to generate and plot samples of (U1, U2) from copulas.
- How to convert uniforms back to original X-scale using empirical quantiles.
- How do I code for Archimedean copulas in R.
- How does the `ecdf(x)(x)` function work.
- How do I make a histogram in R.
- How do I change manual code into a loop.

References

Dewick, PR & Liu, S 2022, 'Copula modelling to analyse financial data', *Journal of Risk and Financial Management*, vol. 15, no. 3, pp. 1–11, doi:10.3390/jrfm15030104.

Statistics Solutions *Kendall's Tau and Spearman's rank correlation coefficient*. Available at: <https://www.statisticssolutions.com/free-resources/directory-of-statistical-analyses/kendalls-tau-and-spearman-rank-correlation-coefficient/> (Accessed: 2 August 2025).

R Code

```
library(quantmod)
library(tidyverse)
library(ggplot2)
library(VineCopula)
RNGversion("4.3.0")

start_date <- as.Date("2010-01-01")
end_date <- as.Date("2025-06-01")
tickers <- c("NVDA", "AMD")
getSymbols(tickers, src = "yahoo", from = start_date, to = end_date, auto.assign = TRUE)
Adj_NVDA <- Ad(NVDA)
Adj_AMD <- Ad(AMD)
X1 <- (log(Adj_NVDA) - log(lag(Adj_NVDA, 1)))[-1]
X1 <- coredata(X1)
X2 <- (log(Adj_AMD) - log(lag(Adj_AMD, 1)))[-1]
X2 <- coredata(X2)
set.seed(2025)
n <- 10000

## Question 1
set.seed(2025)
L1 <- (1 - exp(X1))
L2 <- (1 - exp(X2))
L_empirical <- 0.5*(L1 + L2)

alpha <- c(0.95, 0.99, 0.995)

VaR <- function(losses, alpha) {
  quantile(losses, probs = alpha)
}

ES <- function(losses, alpha) {
  sapply(alpha, function(a) {
    VaR_a <- quantile(losses, probs = a)
    mean(losses[losses >= VaR_a])
  })
}

VaR_L1 <- VaR(L1, alpha)
VaR_L2 <- VaR(L2, alpha)

ES_L1 <- ES(L1, alpha)
ES_L2 <- ES(L2, alpha)

results <- tibble(
  Alpha = alpha,
  VaR_NVDA = VaR_L1,
  VaR_AMD = VaR_L2,
  ES_NVDA = ES_L1,
  ES_AMD = ES_L2
)

print(results)

VaR_empirical <- VaR(L_empirical, alpha)
ES_empirical <- ES(L_empirical, alpha)

VaR_empirical
ES_empirical

## Question 2
set.seed(2025)
n <- 10000

# Step 1: Generate independent uniforms (independence copula)
U1_ind <- runif(n)
U2_ind <- runif(n)
```

```

# Step 2: Transform uniforms to X1 and X2 using inverse empirical CDF
X1_star <- quantile(X1, probs = U1_ind, type = 1)
X2_star <- quantile(X2, probs = U2_ind, type = 1)

# Step 2: Compute simulated losses under independence
L1_star <- 1 - exp(X1_star)
L2_star <- 1 - exp(X2_star)
L_star <- 0.5 * (L1_star + L2_star)

# Step 3: Define alpha
alpha <- c(0.95, 0.99, 0.995)

# Step 4: Define VaR and ES functions
VaR_func <- function(losses, alpha) {
  quantile(losses, probs = alpha)
}

ES_func <- function(losses, alpha) {
  sapply(alpha, function(a) {
    VaR_a <- quantile(losses, probs = a)
    mean(losses[losses >= VaR_a])
  })
}

# Step 5: Compute VaR and ES
VaR_independece <- VaR_func(L_star, alpha)
ES_independece <- ES_func(L_star, alpha)

results_indep <- tibble(
  Alpha = alpha,
  VaR_independece = VaR_independece,
  ES_independece = ES_independece
)

print(results_indep)

## Question 3
set.seed(2025)

# Use previously computed L1 and L2
kendall_tau_manual <- function(x, y, verbose = TRUE) {
  concordant <- 0
  discordant <- 0
  n <- length(x)

  for (i in 1:(n - 1)) {
    for (j in (i + 1):n) {
      dx <- x[i] - x[j]
      dy <- y[i] - y[j]
      prod <- dx * dy
      if (prod > 0) {
        concordant <- concordant + 1
      } else if (prod < 0) {
        discordant <- discordant + 1
      }
    }
  }

  total_pairs <- 0.5 * n * (n - 1)
  tau <- (concordant - discordant) / total_pairs
  prob_concordant <- concordant / total_pairs
  prob_discordant <- discordant / total_pairs

  if (verbose) {
    cat("Concordant pairs:", concordant, "\n")
    cat("Discordant pairs:", discordant, "\n")
    cat("Total pairs:", total_pairs, "\n")
    cat("P(Concordant):", round(prob_concordant, 4), "\n")
    cat("P(Discordant):", round(prob_discordant, 4), "\n")
    cat("Kendall's Tau:", round(tau, 4), "\n")
  }
}

```

```

    return(tau)
}

combined_loss_data <- na.omit(cbind(L1,L2))
n_obs <- nrow(combined_loss_data)
sample_idx <- sample(1:n_obs, size = n_obs)
L1_sub <- L1[sample_idx]
L2_sub <- L2[sample_idx]
kendall_tau <- kendall_tau_manual(L1_sub, L2_sub)

### 2. Spearman's rho using CDF correlation
# (ECDF)
U1_emp <- ecdf(L1)(L1)
U2_emp <- ecdf(L2)(L2)

# Spearman's rho = Pearson correlation between CDF values
spearman_rho <- cor(U1_emp, U2_emp, method = "pearson")

tibble(
  `Kendall Tau (empirical)` = kendall_tau,
  `Spearman Rho (empirical)` = spearman_rho
)

## Question 4
n <- 10000
set.seed(2025)
# Step 1: Create pseudo-observations U1 and U2
U1 <- runif(n)
U2 <- runif(n)

# Step 2: Convert to Z-space using inverse normal CDF
Z1 <- qnorm(U1)
Z2 <- qnorm(U2)

# Step 3: Estimate correlation (Pearson)
rho_hat <- cor(X1, X2, method = "pearson")

# Step 4: Simulate from bivariate normal with estimated rho
Y1 <- Z1
Y2 <- rho_hat * Z1 + sqrt(1 - rho_hat^2) * Z2

U1_gaus <- pnorm(Y1)
U2_gaus <- pnorm(Y2)

# Step 6: Sklar's Theorem - map Y1 and Y2 through empirical quantiles
X1_gaus <- quantile(X1, U1_gaus)
X2_gaus <- quantile(X2, U2_gaus)

# Step 7: Simulate portfolio loss
L1_gaus <- 1 - exp(X1_gaus)
L2_gaus <- 1 - exp(X2_gaus)
L_gaus <- 0.5 * (L1_gaus + L2_gaus)

# Step 8: Compute VaR and ES
alpha <- c(0.95, 0.99, 0.995)

VaR_func <- function(losses, alpha) {
  quantile(losses, probs = alpha, na.rm = TRUE)
}

ES_func <- function(losses, alpha) {
  sapply(alpha, function(a) {
    VaR_a <- quantile(losses, probs = a)
    mean(losses[losses >= VaR_a], na.rm = TRUE)
  })
}

VaR_Gaussian <- VaR_func(L_gaus, alpha)
ES_Gaussian <- ES_func(L_gaus, alpha)

```

```

results <- tibble(
  Alpha = alpha,
  VaR = VaR_Gaussian,
  ES = ES_Gaussian
)

print(results)

ggplot(data.frame(U1_gaus, U2_gaus), aes(x = U1_gaus, y = U2_gaus)) +
  geom_point(alpha = 0.25, size = 0.5) +
  labs(title = "Simulated (U1, U2) from Gaussian Copula",
       x = "U1", y = "U2") +
  theme_minimal()

## Plot the empirical scatterplot
# Convert losses to empirical uniform margins (pseudo-observations)
U1 <- ecdf(X1)(X1) # ECDF-based uniform [0,1]
U2 <- ecdf(X2)(X2)

# Create data frame
empirical_copula_data <- tibble(U1 = U1, U2 = U2)

# Scatterplot of (U1, U2)
ggplot(empirical_copula_data, aes(x = U1, y = U2)) +
  geom_point(alpha = 0.25, size = 0.4, color = "black") +
  labs(
    title = "Empirical Copula Scatterplot of (U1, U2)",
    x = "U1",
    y = "U2"
  ) +
  theme_minimal()

## Question 5
set.seed(2025)
n <- 10000
alpha <- c(0.95, 0.99, 0.995)

# Combine log returns for empirical mapping
x <- cbind(X1, X2)
rho <- as.numeric(cor(X1, X2, method = "pearson"))

simulate_t_samples <- function(df, rho, n, x) {
  # Generate independent uniforms and convert to standard normals
  Ua <- runif(n)
  Ub <- runif(n)
  Za <- qnorm(Ua)
  Zb <- qnorm(Ub)

  # Inverse-gamma scaling for t-distribution
  scale <- 1 / rgamma(n, shape = df / 2, rate = df / 2)
  sqrt_scale <- sqrt(scale)

  # Apply correlation using Cholesky-like construction
  Y1 <- sqrt_scale * Za
  Y2 <- sqrt_scale * (rho * Za + sqrt(1 - rho^2) * Zb)

  # Transform back to uniforms via t-CDF
  U1t <- pt(Y1, df = df)
  U2t <- pt(Y2, df = df)

  # Map to empirical marginals
  X1_t <- unname(quantile(x[,1], probs = U1t, type = 1))
  X2_t <- unname(quantile(x[,2], probs = U2t, type = 1))

  # Compute loss
  L <- 0.5 * ((1 - exp(X1_t)) + (1 - exp(X2_t)))

  # VaR and ES
  VaR <- quantile(L, probs = alpha, type = 1)
  ES <- sapply(VaR, function(v) mean(L[L > v]))

```

```

# Return table
tibble(
  Alpha = alpha,
  nu = df,
  VaR = VaR,
  ES = ES,
  U1t = list(U1t),
  U2t = list(U2t)
)
}

# Apply to all df values without a loop
results_all <- bind_rows(
  simulate_t_samples(3, rho, n, x),
  simulate_t_samples(10, rho, n, x),
  simulate_t_samples(10000, rho, n, x)
)

# View result table
print(results_all %>% select(Alpha, nu, VaR, ES))

# --- Plotting Copula Samples ---

# Extract U1 and U2 for plotting
plot_data <- results_all %>%
  select(nu, U1t, U2t) %>%
  rowwise() %>%
  mutate(data = list(tibble(U1 = U1t[[1]], U2 = U2t[[1]]))) %>%
  select(nu, data) %>%
  unnest(data) %>%
  mutate(nu_label = paste0("v = ", nu))

# Plot
ggplot(plot_data, aes(x = U1, y = U2)) +
  geom_point(alpha = 0.25, size = 0.5) +
  facet_wrap(~ nu_label) +
  labs(title = "t-Copula Samples (U1, U2)",
       x = "U1", y = "U2") +
  theme_minimal()

## Question 6
set.seed(2025)
# --- COMONOTONIC ---
U <- runif(n)
U1_como <- U
U2_como <- U

X1_como <- quantile(X1, probs = U1_como, type = 1)
X2_como <- quantile(X2, probs = U2_como, type = 1)

L1_como <- 1 - exp(X1_como)
L2_como <- 1 - exp(X2_como)
L_como <- 0.5 * (L1_como + L2_como)

VaR_func <- function(losses, alpha) {
  quantile(losses, probs = alpha, na.rm = TRUE)
}

ES_func <- function(losses, alpha) {
  sapply(alpha, function(a) {
    VaR_a <- quantile(losses, probs = a)
    mean(losses[losses >= VaR_a], na.rm = TRUE)
  })
}

VaR_como <- VaR_func(L_como, alpha)
ES_como <- ES_func(L_como, alpha)

# --- COUNTERMONOTONIC ---
set.seed(2025)
U1_counter <- U

```

```

U2_counter <- 1-U

X1_counter <- quantile(X1, probs = U1_counter, type = 1)
X2_counter <- quantile(X2, probs = U2_counter, type = 1)

L1_counter <- 1 - exp(X1_counter)
L2_counter <- 1 - exp(X2_counter)
L_counter <- 0.5 * (L1_counter + L2_counter)

VaR_func <- function(losses, alpha) {
  quantile(losses, probs = alpha, na.rm = TRUE)
}

ES_func <- function(losses, alpha) {
  sapply(alpha, function(a) {
    VaR_a <- quantile(losses, probs = a)
    mean(losses[losses >= VaR_a], na.rm = TRUE)
  })
}

VaR_counter <- VaR_func(L_counter, alpha)
ES_counter <- ES_func(L_counter, alpha)

# Combine results
results <- tibble(
  Alpha = alpha,
  VaR_Comonotonic = VaR_como,
  ES_Comonotonic = ES_como,
  VaR_Countermonotonic = VaR_counter,
  ES_Countermonotonic = ES_counter
)

print(results)

## Question 7
set.seed(2025)
alpha <- c(0.95, 0.99, 0.995)
nu_vals <- c(3, 10, 10000)

# --- Function to estimate conditional probabilities ---
estimate_conditional <- function(U1, U2, alpha_levels) {
  sapply(alpha_levels, function(alpha) {
    num <- sum(U1 > alpha & U2 > alpha)
    denom <- sum(U2 > alpha)
    if (denom == 0) return(NA) else return(num / denom)
  })
}

# --- Gaussian copula ---
gauss_cond <- estimate_conditional(U1_gaus, U2_gaus, alpha)

# --- Independence copula ---
indep_cond <- estimate_conditional(U1_ind, U2_ind, alpha)

# --- Comonotonic copula ---
como_cond <- estimate_conditional(U1_como, U2_como, alpha)

# --- Countermonotonic copula ---
counter_cond <- estimate_conditional(U1_counter, U2_counter, alpha)

# --- Initialize result tibble ---
cond_df <- tibble(
  Alpha = alpha,
  Independence = indep_cond,
  Gaussian = gauss_cond,
  Comonotonic = como_cond,
  Countermonotonic = counter_cond
)

# --- Extract t-copula conditional probabilities by filtering ---
for (nu in nu_vals) {

```

```

# Filter only rows with the given df (nu)
row <- results_all %>% filter(nu == !!nu)

# Extract U1 and U2 from the list-columns
U1 <- row$U1t[[1]]
U2 <- row$U2t[[1]]

# Estimate conditional probabilities and add to tibble
cond_df[[paste0("t_df_", nu)]] <- estimate_conditional(U1, U2, alpha)
}

# View final table
print(cond_df)

# Combine all results into one tidy data frame for ggplot
summary_long <- bind_rows(
  tibble(Alpha = alpha, VaR = VaR_empirical, ES = ES_empirical, Copula = "Empirical"),
  tibble(Alpha = alpha, VaR = VaR_L, ES = ES_L, Copula = "Independence"),
  tibble(Alpha = alpha, VaR = VaR_Gaussian, ES = ES_Gaussian, Copula = "Gaussian"),
  tibble(Alpha = alpha, VaR = results_list$`3`$VaR, ES = results_list$`3`$ES, Copula = "t
(v=3)"),
  tibble(Alpha = alpha, VaR = results_list$`10`$VaR, ES = results_list$`10`$ES, Copula = "t
(v=10)"),
  tibble(Alpha = alpha, VaR = results_list$`10000`$VaR, ES = results_list$`10000`$ES, Copula
= "t (v=10000)"),
  tibble(Alpha = alpha, VaR = VaR_como, ES = ES_como, Copula = "Comonotonic"),
  tibble(Alpha = alpha, VaR = VaR_counter, ES = ES_counter, Copula = "Countermonotonic")
)

print(summary_long, n=28)

# Plot VaR
ggplot(summary_long, aes(x = Alpha, y = VaR, color = Copula, group = Copula)) +
  geom_line(size = 1.2) +
  geom_point(size = 2) +
  labs(title = "VaR $\alpha$ (L) Comparison across Copulas", y = "Value at Risk (VaR)", x = "Alpha")
+
  theme_minimal() +
  theme(legend.title = element_blank())

# Plot ES
ggplot(summary_long, aes(x = Alpha, y = ES, color = Copula, group = Copula)) +
  geom_line(size = 1.2) +
  geom_point(size = 2) +
  labs(title = "ES $\alpha$ (L) Comparison across Copulas", y = "Expected Shortfall (ES)", x =
"Alpha") +
  theme_minimal() +
  theme(legend.title = element_blank())

## Question 8
set.seed(2025)
# Convert to pseudo-observations (ranks converted to percentiles)
U1 <- ecdf(X1)(X1)
U2 <- ecdf(X2)(X2)

# --- Step (a): Fit best Archimedean copula ---
copula_fit <- BiCopSelect(U1, U2, familyset = 1:10)

# Print summary of selected copula
print(copula_fit)
summary(copula_fit)

# VineCopula t
t_fit <- BiCopSelect(U1, U2, familyset = 2)

# Print summary of selected copula
print(t_fit)
summary(t_fit)

# --- Step (b): Interpret copula parameters ---
cat("Fitted copula family:", copula_fit$family, "\n")

```

```

cat("Copula name:", BiCopName(copula_fit$family), "\n")
cat("Parameter(s):", copula_fit$par, copula_fit$par2, "\n")

# Extract lower and upper tail dependence
tail_dep <- BiCopPar2TailDep(copula_fit)
cat("Tail dependence (lower, upper):", tail_dep$lower, tail_dep$upper, "\n")

# --- Step (c): Simulate losses from selected copula ---
set.seed(2025)

# Simulate 10,000 pseudo-observations (U1*, U2*) from the fitted copula
U_sim <- BiCopSim(10000, copula_fit)

# Transform back to empirical scale using inverse ECDFs
X1_sim <- quantile(X1, probs = U_sim[, 1], type = 1)
X2_sim <- quantile(X2, probs = U_sim[, 2], type = 1)

# Convert to simulated losses
L1_sim <- 1 - exp(X1_sim)
L2_sim <- 1 - exp(X2_sim)

# --- Step (d): Conditional vs Unconditional Histograms ---
alpha <- 0.995
VaR_L2 <- quantile(L2_sim, probs = alpha)

# Subset for conditional distribution: L1 | L2 > VaR
L1_cond <- L1_sim[L2_sim > VaR_L2]

# Create dataframe as before
df_plot <- data.frame(
  Loss = c(L1_sim, L1_cond),
  Type = rep(c("Unconditional", paste0("Conditional | L2 > VaR_", alpha)),
    times = c(length(L1_sim), length(L1_cond)))
)

# Plot with facets (separate panels)
ggplot(df_plot, aes(x = Loss, fill = Type)) +
  geom_histogram(bins = 50, color = "black", alpha = 0.7) +
  facet_wrap(~ Type, ncol = 2, scales = "free_y") +
  labs(title = "Histograms of Unconditional and Conditional L1 Losses",
    x = "Simulated Loss (L1)", y = "Frequency") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none")

## Question 9
# Vary weights and compute risk metrics ---
alpha <- 0.99
w_grid <- seq(0, 1, by = 0.0001)

weight_metrics <- tibble(
  w = w_grid,
  VaR = NA_real_,
  ES = NA_real_,
  Sharpe = NA_real_,
  ES_to_VaR = NA_real_
)

for (i in seq_along(w_grid)) {
  w <- w_grid[i]
  L_w <- w * L1_sim + (1 - w) * L2_sim

  VaR_w <- quantile(L_w, probs = alpha)
  ES_w <- mean(L_w[L_w >= VaR_w])

  weight_metrics$VaR[i] <- VaR_w
  weight_metrics$ES[i] <- ES_w
  weight_metrics$Sharpe[i] <- -mean(L_w) / sd(L_w)
  weight_metrics$ES_to_VaR[i] <- ES_w / VaR_w
}

# Extract optimal weights ---

```

```

opt_w_ES      <- weight_metrics$w[which.min(weight_metrics$ES)]
opt_w_VaR     <- weight_metrics$w[which.min(weight_metrics$VaR)]
opt_w_Sharpe  <- weight_metrics$w[which.max(weight_metrics$Sharpe)]
opt_w_ratio   <- weight_metrics$w[which.min(weight_metrics$ES_to_VaR)]

final_weights <- tibble(
  Method = c("ES-minimising", "VaR-minimising", "Sharpe-maximising", "ES:VaR-minimising"),
  Optimal_w = round(c(opt_w_ES, opt_w_VaR, opt_w_Sharpe, opt_w_ratio), 3)
)

final_weights

# --- Step 8: Plot risk metrics ---

ggplot(weight_metrics, aes(x = w, y = ES)) +
  geom_line(color = "red") +
  labs(title = "Expected Shortfall (ES) vs Portfolio Weight", x = "Weight on NVDA", y =
"ES") +
  theme_minimal()

ggplot(weight_metrics, aes(x = w, y = Sharpe)) +
  geom_line(color = "blue") +
  labs(title = "Sharpe Ratio vs Portfolio Weight", x = "Weight on NVDA", y = "Sharpe Ratio")
+
  theme_minimal()

print(final_weights)

```